



Project Title: CivicPulse – Empowering Citizens to Improve Their City



Problem Statement



- Citizens face difficulty reporting civic issues like potholes, garbage, broken streetlights, etc.
- No centralized digital platform for transparent tracking.
- Lack of communication between citizens and authorities.
- Issues are not prioritized based on community importance.
- Manual complaint systems are slow and inefficient.



Proposed Solution



We propose CivicPulse, a full-stack web application that:

- Allows citizens to register and log in securely
- Enables users to report civic issues with details and images
- Lets community members like and comment on issues
- Prioritizes issues based on community votes
- Provides an Admin Dashboard for authorities to manage issues
- Ensures secure role-based access control



System Architecture



1. Frontend (Client Side)

- Built using React.js
- Displays UI (Login, Register, Dashboard, Browse Issues)
- Sends requests to backend using Axios
- Stores JWT token after login

2. Backend (Server Side)

- Built using Node.js & Express.js
- Handles API requests
- Manages authentication (JWT)
- Applies role-based access (User / Admin)
- Communicates with database

3. Database (MongoDB Atlas)

- Stores user details
- Stores issues data
- Stores likes and comments
- Stores user roles

User → React Frontend → Express Backend → MongoDB Database → Response back to User



TECH STACK



Engineer to Excel

Frontend:

- React.js
- React Router DOM
- Axios
- Tailwind CSS

Backend:

- Node.js
- Express.js
- JWT (Authentication)
- bcrypt.js (Password hashing)

Database:

- MongoDB Atlas
- Mongoose ODM

Security:

- Role-Based Access Control
- Protected Routes Middleware



IMPLEMENTATION PLAN



Engineer to Excel

Phase 1 – User Module

- Registration & Login system
- JWT authentication
- Issue creation
- Browse issues
- Like & Comment functionality

Phase 2 – Admin Module

- Admin Dashboard UI
- Analytics cards
- Issue status management
- Delete issues

Phase 3 – Backend Protection

- Role-based middleware
- Admin-only routes
- Secure API validation

Phase 4 – Future Enhancements

- Real-time notifications
- Image upload storage
- Geo-location tracking
- AI-based issue categorization



CONCLUSION



CivicPulse successfully:

- Provides a secure civic issue reporting platform
- Enables community engagement through likes & comments
- Implements role-based access control
- Uses scalable MERN architecture
- Improves transparency between citizens and authorities

--CivicPulse promotes digital democracy and responsible civic engagement.

THANK YOU